Domain Adversarial Reinforcement Learning

Bonnie Li
McGill University, Mila

Vincent François-Lavet
VU Amsterdam

Thang Doan
McGill University, Mila

Joelle Pineau
McGill University, Mila

Abstract

We consider the problem of generalization in reinforcement learning where visual aspects of the observations might differ, e.g. when there are different backgrounds. This class of problems is referred to as block MDPs and we assume that our agent has access to only a few of the MDPs from the block MDP distribution during training. The performance of the agent is then reported on new unknown test domains taken from the distribution (e.g. unseen backgrounds). For this "zero-shot RL" task, we enforce the learning of a distribution of representations that is invariant to the specific training domains via a domain adversarial component that modifies the weights of a shared encoder. We empirically show that it allows achieving a significant generalization improvement to new unseen domains.

1 Introduction

Deep reinforcement learning (deep RL) has achieved great success in solving complex control tasks from high-dimensional pixel inputs, such as video games [Bellemare et al., 2013, Cobbe et al., 2019a] and continuous control tasks [Tassa et al., 2018].

In real world applications such as robotics, interacting with real environment can be expensive and dangerous (no guarantee on the policy safety). As such, simulation is used as a safer and cheaper alternative [Chaffre et al., 2020, Matas et al., 2018]. But RL agents can overfit very easily to the enviroment in which they were trained [Cobbe et al., 2019b, Zhang et al., 2018b]. It remains a challenge for agents to generalize to environments beyond which they were trained in. For the same task dynamics, RL agents can be prone to brittleness caused by noise injection in the observation space, distractors [Ding et al., 2020], and background change [Zhang et al., 2018a, 2020b].

Learning a robust representation is then key towards generalization in deep RL. The representation should retain sufficient task-relevant information in order to learn a good policy, while ignoring irrelevant features such as fine grained details of the environemnt. Reconstruction-based methods [Yarats et al., 2019] encourage the representation to caputre all details of the environment and will more likely lead to memorisation and overfitting. Recent methods, capturing spatial [Srinivas et al., 2020], or temporal [Mazoure et al., 2020] generative factors, or bisimilation-relevant factors [Zhang et al., 2020b], aim to learn features that increase sample efficiency of the agent.

In this work, we propose a simple approach that improves generalization to all visual domains with the same task dynamics. Our approach aims to learn an abstract representation space at the output of an encoder that (i) captures the important features from the source domains and (ii) is invariant with respect to the domain distribution shift. We evaluate our approach on DeepMind Control tasks, and examine zero-shot generalization to unseen visual domains with both stationary and non-stationary backgrounds.

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2 Related Works

2.1 Feature Distribution Alignment

Our work builds on early work in feature distribution alignment. Aligning distributions in the feature space can be done by optimizing some measure of discrepancy, such as the maximum mean discrepancy (MMD). Another popular approach for aligning distributions is via an adversarial discriminator that distinguishes between the feature distributions. This was used in the setting of unsupervised domain adaption, where labeled training data is available on a source domain, but the goal is to have good performance on a target domain with only unlabeled data. By minimizing the domain classification accuracy of the adversarial discriminator, one can align the feature representations across source domain and target domain (Ganin and Lempitsky [2014], Tzeng et al. [2017]). It is however unclear whether the features learned in this setting can zero-shot transfer to unseen domains.

2.2 Visual Transfer in Reinforcement Learning

Reinforcement learning environments with different visual backgrounds can be formalized as a block MDP. Several recent works examined varied instances of generalization and tranfer within these environments. Robust Domain Randomization (RDR, Slaoui et al. [2020]) considers the problem of visual randomization, and attempts to zero-shot transfer to new domain by minimizing the Euclidean distance between feature representations across randomization of training domains. MISA (Zhang et al. [2020a]) uses tools from causal inference, namely ICP and IRM, to learn Model-Irrelevance State Abstractions, which shows generalization on low-dimensional RL tasks and visual imitation learning. PAD (Hansen et al. [2020]) employs a self-supervised inverse dynamic model combined with data augmentation for faster adaption during deployment when the policy is deployed in a different domain. WAPPO (Roy and Konidaris [2020]) examines the problem of unsupervised domain adaption in the RL setting, and uses Wasserstein GAN to measure and align the feature distributions from the source and target domain. While WAPPO considers one source and one target domain, and has access to target domain observations during training, in our setting, we aim to zero-shot generalize to unseen domains from training on a distribution of environments.

3 Preliminaries/Background

3.1 Reinforcement learning

We consider Reinforcement Learning (RL) in the standard Markov Decision Process (MDP) $\mathcal{M} := \langle \mathcal{S}, \mathcal{A}, r, P, S_0, \gamma \rangle$, where $\mathcal{S}$ is the state space; $\mathcal{A}$ is the action space; given states s, $s' \in \mathcal{S}$, action $a \in \mathcal{A}$, $P(s'|s,a)$ is the transition probability of transferring s to s' under the action a and $r(s,a)$ is the reward collected at state s after executing the action a; $S_0 \subseteq \mathcal{S}$ is the set of initial states and γ is the discount factor.

The agent executes actions in its RL environment according to some policies. We define a stochastic policy $\pi : \mathcal{S} \times \mathcal{A} \to [0,1]$ such that $\pi(a|s)$ is the conditional probability that the agent takes action $a \in \mathcal{A}$ after observing the state $s \in \mathcal{S}$.

We define the state value function V_π and the state-action value function Q_π as:

$$V_\pi(s) = \mathbb{E}_\pi \left[\sum_{k=t}^{\infty} \gamma^{k-t} r(s_k, a_k) \middle| s_t = s \right],$$

$$Q_\pi(s,a) = \mathbb{E}_\pi \left[\sum_{k=t}^{\infty} \gamma^{k-t} r(s_k, a_k) \middle| s_t = s, a_t = a \right].$$

The objective is to find a policy π which maximizes the expected discounted reward V_π. The gap between Q_π and V_π is known as the advantage function:

$$A_\pi(s,a) = Q_\pi(s,a) - V_\pi(s).$$

3.2 Block MDPs

Our problem set up of reinforcement learning environments with different visual backgrounds can be formalized as a block MDP, defined by $\mathcal{M} := \langle \mathcal{S}, \mathcal{A}, \mathcal{X}, p, q, R, \gamma \rangle$. Such block MDP defines a distributions of MDPs. This setting assumes that the related MDPs would share the same unobserved state space $\mathcal{S}$, action space $\mathcal{A}$, latent transition distribution $p(s' \mid s, a)$, and reward function R. But the MDPs can differ in their observed state space $\mathcal{X}$ (also formalized as the context space [Du et al., 2019]), and emission function $q(x \mid s)$, which maps the hidden state space to the observed context space.

We make a few relaxations and assumptions from the original block structure assumption (Du et al. [2019]) for our problem set up. Similarly to Zhang et al. [2020a], we assume that there exists an invariant state embedding from the observed state space, mapped by $\theta_e : \mathcal{X} \mapsto \mathcal{Z}$. We further assume the existance of an invariant dynamic model and reward model within such state embedding. This implies that optimal behaviour is achievable with the same policy across all MDPs within the block.

3.3 Soft Actor Critic

Soft Actor Critic (SAC, Haarnoja et al. [2018]) is an off-policy actor critic method using the maximum entropy framework. It trains a neural network Q_θ to approximate the agent's current Q-function, by minimizing the soft Bellman residual

$$L_Q = \Big(Q(s_t, a_t) - \big(r_t + \gamma \bar{V}(s_{t+1})\big)\Big)^2$$

where $\bar{V}$ is the target value function computed as

$$\bar{V}(s_t) = \bar{Q}(s_t, a_t) - \alpha \ln \pi(a_t \mid s_t)$$

where $\bar{Q}$ is the target Q function, computed as the exponential moving average of Q_θ. The policy is trained by minimizing the divergence between the policy and the exponential of the soft Q function

$$L_\pi = D_{KL}\Big(\pi(\cdot \mid s_t) \parallel exp(\frac{1}{\alpha} Q(s_t, \cdot))\Big)$$

Several recent works (Yarats et al. [2019], Srinivas et al. [2020], Laskin et al. [2020], Kostrikov et al. [2020]) use representation loss and data augmentation to improve SAC from pixels. Data augmentation, namely random crop alone, was shown to be particularly effective. Following RAD Laskin et al. [2020], in our setting we build on SAC with randomly cropped input pixel observations.

4 Domain Adversarial Reinforcement Learning

We seek to train an agent that maximizes its expected cumulative reward over a distribution of MDPs defined by a block MDP. That is, the agent should not only maximize rewards on the training environments, but should also generalize well to unseen environments drawn from the same block without any additional training. To do so, we propose to combine a strong model-free RL algorithm, SAC (with data augmentation), with a proposed auxillary adversarial loss to improve generalization.

We aim to learn robust representations from high-dimensional observations that can directly generalize to diverse MDP contexts drawn from the same block, ie. different visual setting. Here a good representation would be one that is informative enough for good task performance, and is general enough to generalize zero-shot to unseen visual settings.

Intuively, if we learn representations that are invariant to background information, the representations can then generalize to unseen backgrounds. To explicitly remove the backgrounds, we propose to use an adversarial discriminator to measure and align feature distributions across domains. We use the adversarial discriminator to predict the context of the MDP (ie. label of the background in our setting), and try to force the features to be as indistinguishable as possible for the discriminator via a gradient reversal layer. See figure 1 for the proposed architecture. We now describe our approach in detail.

We use an encoder $E : \mathcal{X} \mapsto \mathcal{Z}$, parameterized by θ_e, to map the high dimensional states into lower-dimensional representatios. We use the resulting feature vectors z for the policy π and the

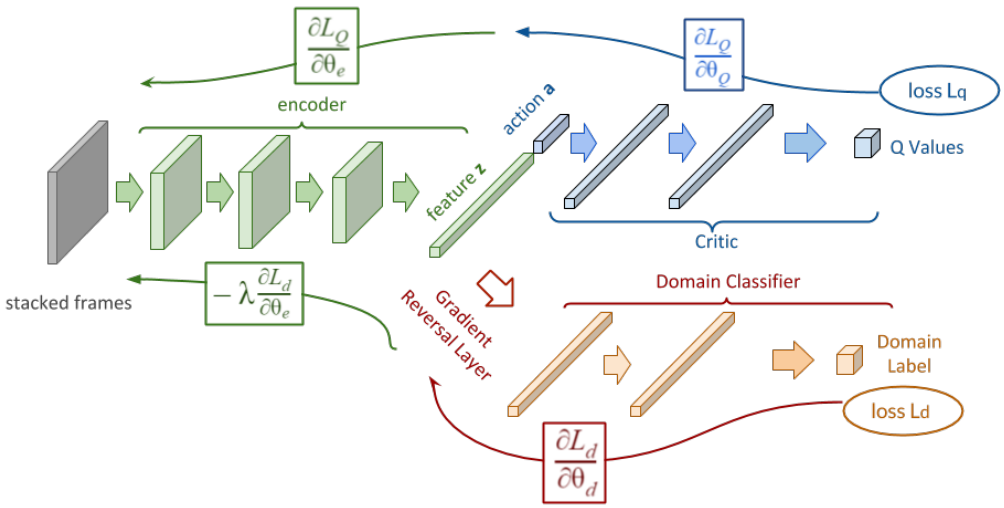

Figure 1: The proposed architecture builds a domain classification module (red) along with the critic and encoder of SAC. Feature distributions across domains are aligned via the gradient reversal layer between the encoder and the domain classifier. The gradient reversal layer multiplies the gradient by a certain negative constant during back propagation. Otherwise, the training follows the standard procedure and minimizes the critic loss and domain classification loss. Gradient reversal ensures that features across domains are indistinguisable for the domain classifier, which encourages domain invariant features.

critic Q, parametrized by θ_π and θ_Q respectively. Building on SAC, the policy is trained with the encoder detached, i.e. the policy loss will not backpropagate to the encoder. The encoder is trained jointly with the critic, for which we concatenate the state features with actions as inputs. Together, we have

$$\forall s \in \mathcal{S}, \quad z = E_\theta(s), \; Q(s,a) = Q_\theta(z,a), \; a \sim \pi_\theta(a \mid \bar{z}),$$

where s is the high dimensional state, z is the encoded state feature, and $\bar{z}$ denotes state feature detached from encoder.

Adversarial Discriminator. The adversarial discriminator, in our setting, is the domain classifier D_θ predicting the domain label of the state from state features. This is trained with the domain classification loss $L_D(s,y)$, which is the negative log-likelihood of the correct domain label output.

$$L_D(s,y) = -\ln P_D(y \mid z), z = E_\theta(s) \; ,$$

where z is the feature of state s from the encoder E_θ, and y is the correct domain label of state s.

Gradient Reversal Layer. Following Domain Adversarial Neural Network (DANN, Ganin and Lempitsky [2014]), we connect the adversarial discriminator with the encoder via a gradient reversal layer (GRL). Mathematically, we can formalize the gradient reversal layer as a "pseudo function" G_λ characterized by the following two (incompatible) equations:

$$G_\lambda(z) = z \; , \; \frac{dG_\lambda}{dz} = -\lambda I \; ,$$

where I is the identity matrix. During forward pass, GRL acts as an identity transform. During backpropagation though, GRL takes the gradient from the subsequent level, multiplies it by a pre-defined negative constant $-\lambda$ and passes it to the preceding layer.

Adaption factor λ. The gradient reversal layer has only one meta-parameter λ, which is not updated during backpropagation. This constant scales the gradient during backpropagation. We use this constant to mitigate the effect of noise from the discriminator in the early training stage, by increasing λ from 0 to 1 during the course of training. Similarly to Ganin and Lempitsky [2014], we update λ using the following

$$\lambda_p = \frac{2}{1 + exp(-10 \cdot p)} - 1 \; ,$$

where p is increased linearly from 0 to 1 over the first million environment steps of training, i.e. $p = \text{current step}/1000000$.

With the gradient reversal layer between encoder features and domain classifier, by directly minimizing the domain classification loss via backpropagation, we train the domain classifier D_θ to distinguish the background class from state features, and simultaneously train the encoder E_θ to produce state features that **maximizes** the classification error (by taking the reversed gradient), therefore encouraging invariant features that are indistinguishable across domains.

Overall, we jointly minimize the critic loss and the domain classification loss. Separately, the policy is trained (with the encoder detached) with policy loss $L\pi$ following from SAC in 3.3. To jointly train the encoder, critic, and domain classifier, we minimize

$$L_\theta = L_Q + \beta \cdot L_D \ ,$$

where L_D is the discriminator loss described above, L_Q follows from the Q loss in SAC in 3.3, and β scales the two losses.

5 Experiments

In this section, we empirically examine the effectiveness of our proposed method. In section 5.1, we evaluate our method along with state-of-the-art RL methods on four continuous control tasks. We then examine properties of the latent space learned in 5.1. Finally, we evaluate how the trained model generalizes to non-stationary environments.

5.1 Comparison against baseline

Our experiments are designed to examine the following questions: 1, How well do reinforcement learning algorithms generalize to unseen states with the same underlying structure? 2. How does the proposed auxiliary adverarial objective compare with state of the art methods?

To answer the posed questions, we set up our experiments as follows: we experiment with four tasks from DeepMind Control Suite Tassa et al. [2018] and measure zero shot generalization to unseen environments with identical task dynamics. We train the agent on four training environments, each with a different stationary background. We test the agent on two testing environments with different backgrounds. The training environments and the testing environments share identical underlying dynamics and differ simply in backgroun visualizations.

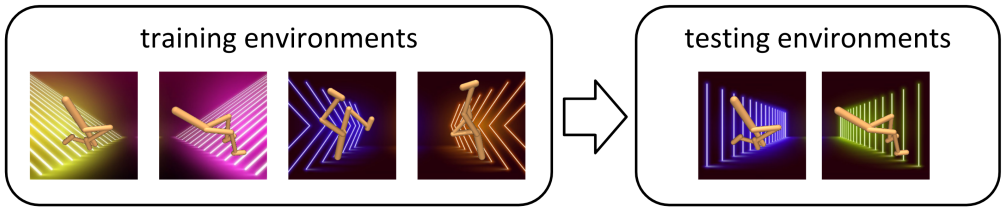

Figure 2: Left four: training environments for the RL agent on deepmind control tasks. Right two: testing environments in which we evaluate the RL agent. Both sets of environments use stationary backgrounds of images, the backgrounds differ simply by shape of lines and colors.

Robustness to observation changes, such as backgorunds, is vital to deploying RL in the real world. We evaluate zero shot generalization of our algorithm on a fixed set of two unseen backgrounds. The backgrounds differ from the ones in the training environment by shape of lines and color. While training on the training set, we evaluate the algorithm's ability to transfer zero-shot to the testing environment.

We evaluate against SAC with data augmentation (RAD, Laskin et al. [2020]), a state-of-the-art model free RL method from images, and upon which we build DARL. Results are shown in figure 3. DARL outperforms SAC on testing environments on all four tasks. The domain adversarial module improves generalization to unseen testing environment, and in some cases also improves the rewards on training environments as it encourages efficient task-relevant representations.

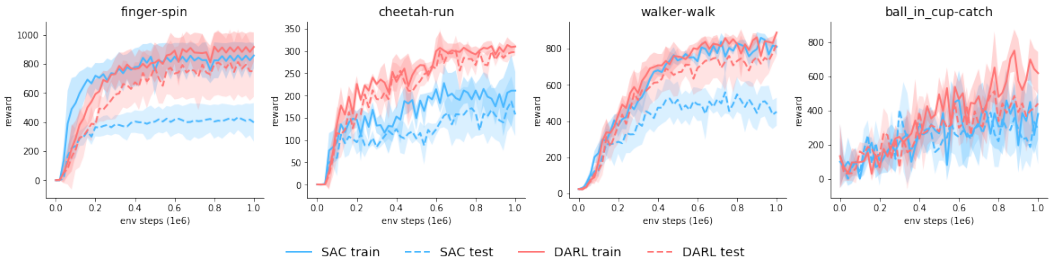

Figure 3: **Deepmind control tasks.** Solid and dashed lines indicate training and testing environments, respectively. DARL outperforms SAC on testing environments across all four tasks. Results are averaged over 3 seeds, line shows the mean and shaded area shows standard deviation.

5.2 DARL learns robust latent representations

To visualize the representations learned by DARL and SAC in the previous section, we use a t-SNE plot (Figure 4). We collect one trajectory (one episode, 500 observations) in each training and testing environment, model the latent representations (50-dimentions) of the observations in a two-dimensional map. While SAC learns different embeddings for different backgrounds, our model outputs the same embedding independently of the background. This confirms that DARL could learn task relevant representations and remove features that are not necessary for the policy.

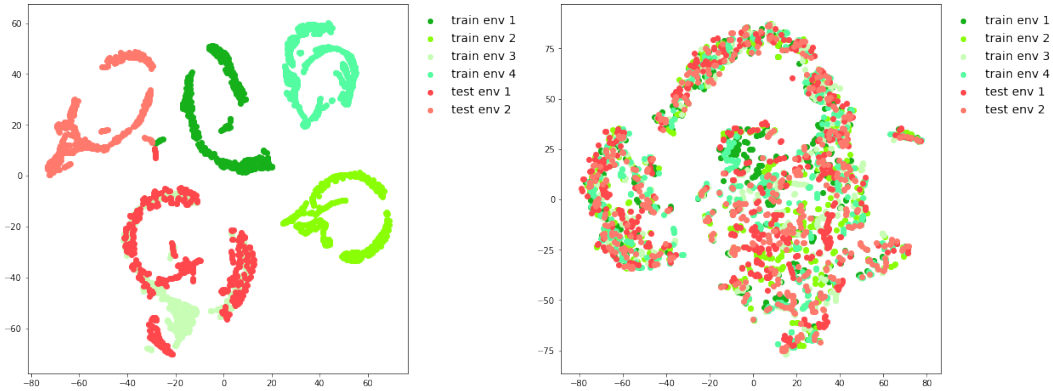

Figure 4: **t-SNE of latent spaces** learned by SAC (left) and by DARL (right). Green shows state representations from training environments, red shows those from unseen testing environments. While SAC learns features that are background dependent, DARL removes those task irrelevant features and learns domain-invariant representations.

5.3 Zero-Shot Generalization to Non-Stationary Domains

In the previous experiment, we show that out method can generalize to test environments with different stationary backgrounds. We further assess generalization ability of our method by addressing the following question: ***Can DARL also zero-shot generalize to non-stationary environments?***

This setting would be more similar to real world environments where the visual setting rarely stays exactly the same. To further test the robustness of our methods, we test on environments with natural video backgrounds (Zhang et al. [2018a]). We use the trained policies from section 5.1, which are trained on stationary image backgrounds, and directly evaluate them on natural video environments.

We use environments from PAD (Hansen et al. [2020]), SAC-AE (Yarats et al. [2019]), and DBC (Zhang et al. [2020b]. We evalute zero-shot generalization of the algorithms trained with the training backgrounds shown in 2, to natural background environments shown in Figure 5, for the cheetah-run task. Results averaged over three environment seeds, dots indicate the mean and error bars show the standard deviation.

Figure 5: **Non-stationary testing environments indexed.** The number indicates index of the environment. Environments all use natural video backgrounds. Environment 1 - 4 uses background videos from [Hansen et al., 2020]; environment 5 uses simple moving distractors in the background, consisted of colored balls moving around and bouncing off the frames, as used in [Yarats et al., 2019], [Zhang et al., 2020b], environment 6 uses natural car driving video from the Kinetics dataset [Kay et al., 2017], as done in [Zhang et al., 2020b].

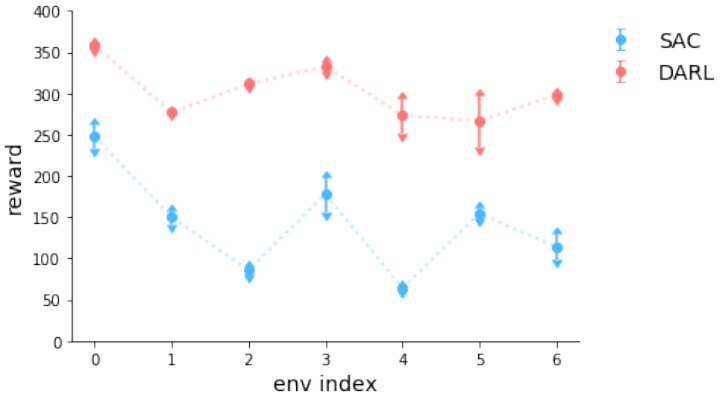

Figure 6: **Generalization Rewards on non-stationary environments from figure 5**. Zero-shot generalization to non-stationary environments of SAC (blue) and DARL (red), on the cheetah-run task. X-axis indicates the environment as shown and indexed in figure 5, 0 indicates original training performance on the training environment with stationary images.

Results are shown in figure 6. SAC's performance drop more drastically from the training environment. It also varies more by environments, which we conjecture since SAC still contain information about the background in its representation, certain visual variations can lead to considerable changes in its policy. Comparatively, DARL is more robust to visual variations and more stable in performance across all environments.

How does the generalization rewards correlate to distance in the latent representation?
A possible measure of dissimilarities for visual environments is the distance between feature distributions in the latent space. To examine the correlation between such measure and generalization, we plot the zero-shot performance with respect to the computed L2 distance of the feature mean in figure 7.

For each natural video environment from figure 5, we compute the mean of feature distributions of its observations (averaged over 10 trajectories), and compute the L2 distance with respect to the mean of the feature distributions from the training environment.

Compare to SAC, DARL aligns these feature distributions from testing environments more close to that of the training environment. Furthermore, the drop in performance of DARL is more moderate and stable than that of SAC. SAC's performance dropped by 74.1% at the testing environment furthest in latent space, while DARL's performance dropped 23.5% in its furthest testing environment. Note that since DARL and SAC each has its own learned feature space and respective mappings, the

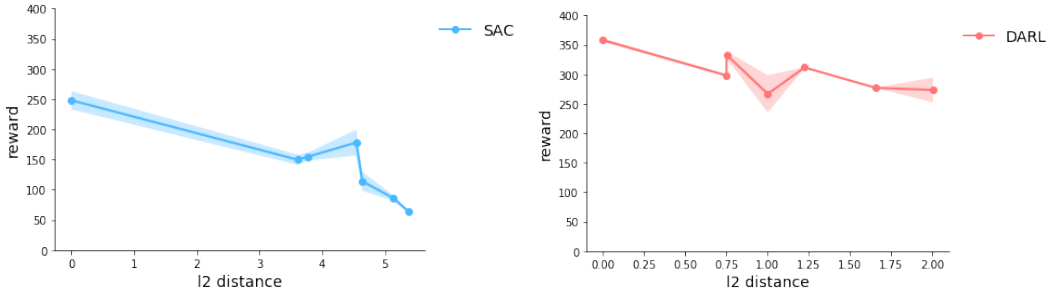

Figure 7: **Generalization rewards w.r.t. dissimilarity of testing environment to training environment**, as measured by L2 distance between feature distributions. Compare to DARL, SAC's performance decrease drastically as the environment moves further from the training environment.

numerical L2 distance in such spaces are not directly comparable. But generally, as observed in figure 7, larger distance between the feature mean corresponds to worse performance.

6 Conclusion

In this paper we present Domain Adversarial Reinforcement Learning (DARL), an adversarial learning approach designed to improve generalization of deep reinforcement learning agents. Compared to state-of-the-art model-free RL from images, DARL shows a significant improvement in zero-shot generalization to unseen domains, both stationary and non-stationary. We hope continued research at the intersection of adversarial learning and reinforcement learning can lead to more robust deep reinforcement learning algorithms.

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